3. $y = \sqrt{(3^x + x)}$

(a) Complete the table below, giving the values of $y$ to 3 decimal places.

| $x$ | 0 | 0.25 | 0.5 | 0.75 | 1 |
|---|---|---|---|---|---|
| $y$ | 1 | 1.251 | | | 2 |

**(2)**

(b) Use the trapezium rule with all the values of $y$ from your table to find an approximation for the value of

$$\int_0^1 \sqrt{(3^x + x)}\, dx$$

You must show clearly how you obtained your answer.

**(4)**

(c) Explain how the trapezium rule could be used to obtain a more accurate estimate for the value of

$$\int_0^1 \sqrt{(3^x + x)}\, dx$$

**(1)**